

CS471 – Lesson 11 Solutions Key

Non-Deterministic Search: Markov Decision Processes (MDPs)

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Non-Determinism & Risk

Question: What does "non-deterministic" mean in the context of an autonomous UAV navigating a contested grid? How is $P(s'|s, a) = T(s, a, s')$ related to non-determinism?

Solution

Non-deterministic means that taking a specific action does not guarantee a specific outcome due to environmental variables (e.g., wind, enemy jamming, mechanical failure). The notation $P(s'|s, a) = T(s, a, s')$ represents the mathematical probability that taking action a in state s will successfully transition the agent into the specific target state s' .

Scenario: A UAV is navigating to an extraction point.

- **Case A:** The living reward is $R(s) = -0.01$ (Fuel is plentiful).
- **Case B:** The living reward is $R(s) = -0.04$ (Fuel is critically low).

Question: Identify and describe how the optimal policy changes (takes more or less risk) when the reward drops to -0.04. Why does the policy take more risk?

Solution

When the living reward drops to -0.04, the penalty for existing in the environment increases significantly. The policy will take much more risk (e.g., flying close to enemy radar or hazardous terrain) to reach the goal state faster. It takes more risk because the accumulated negative reward of a safe, long route becomes mathematically worse than the probability of failing on a fast, risky route.

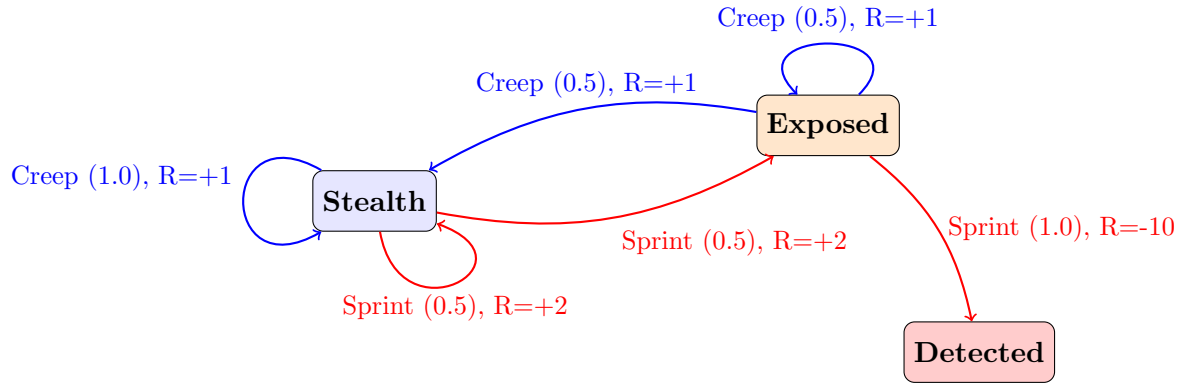
2. The Markov Property

Question: State what this means in plain English. Why is this important for AI?

Solution

In plain English, the Markov Property means that "the future is independent of the past given the present." The outcome of an action depends *only* on the current state the agent is in, not how it arrived there. This is critical for AI because it vastly reduces the state space; the AI does not need to memorize its entire historical path to calculate its next optimal move.

3. MDP Transition & Reward Table



Solution

S	A	S'	$T(S, A, S')$	$R(S, A, S')$
Stealth	Creep	Stealth	1.0	+1
Stealth	Sprint	Stealth	0.5	+2
Stealth	Sprint	Exposed	0.5	+2
Exposed	Creep	Stealth	0.5	+1
Exposed	Creep	Exposed	0.5	+1
Exposed	Sprint	Detected	1.0	-10

Part 2: Student Independent Practice Solutions

Electronic Warfare (EW) Operations

1. Build a table of transition probabilities and rewards for each state.

Solution

S	A	S'	T(S, A, S')	R(S, A, S')
Standby	Loiter	Standby	1.0	+1
Standby	Jam	Standby	0.5	+3
Standby	Jam	Emitting	0.5	+3
Emitting	Loiter	Standby	0.5	+1
Emitting	Loiter	Emitting	0.5	+1
Emitting	Jam	Targeted	1.0	-15

2. Draw the Markov Network for the problem.

